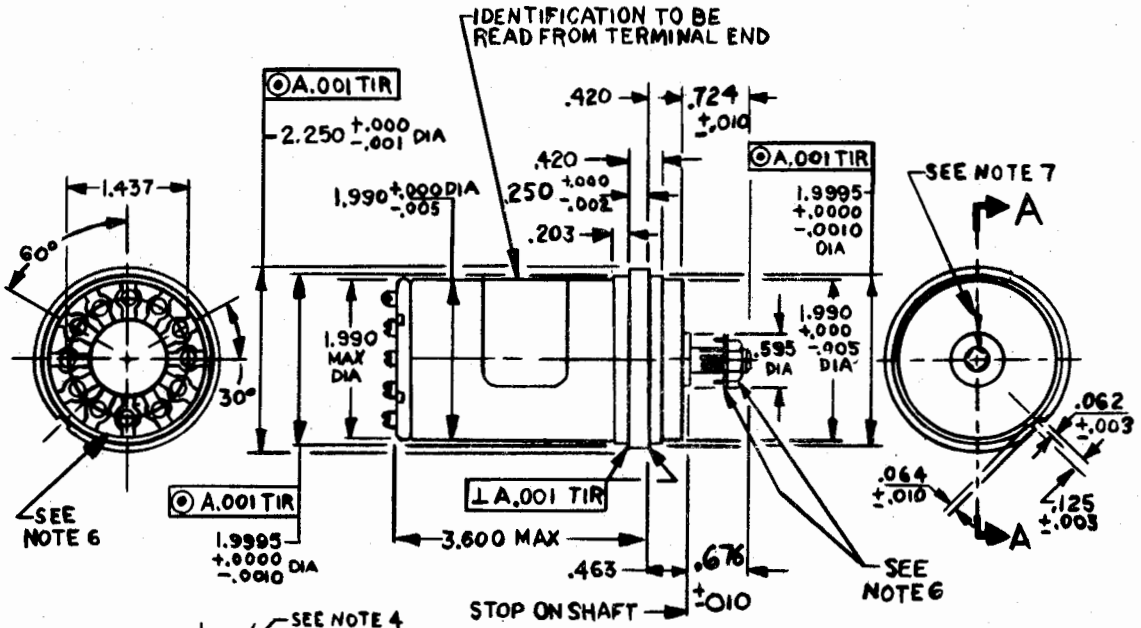
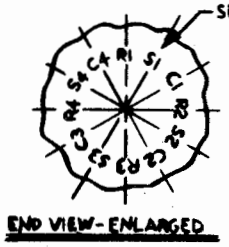
MILITARY SPECIFICATION SHEET

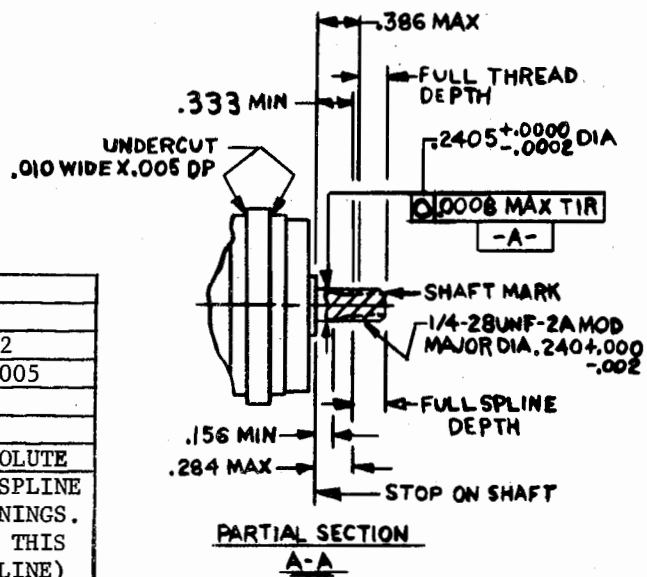
RESOLVER, ELECTRICAL

TYPE 23R32W4a

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of MIL-R-23417.







SPLINE DATA

TEETH	22.
PITCH	96
PITCH DIAMETER	.2291+.000-.002
OUTSIDE DIAMETER	.2405+.0000-.0005
PRESSURE ANGLE	20°
ROOT DIA. MAX	.205
TOOTH FORM	FULL DEPTH INVOLUTE
NOTE: THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS. IT IS NOT TO BE USED AS A GEAR. THIS SPLINE SHALL BE INSPECTED BY (SPLINE) GO COMPOSITE RING GAGE BUORD929471.	

FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are .XXX ± .005 and angles ±2°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequence shall be as shown and identified by molded lettering on the end cap or permanently affixed thereto. The end cap and terminal block shall be secured by screws other than the terminal screws. Mounting screw positions optional. Terminal block designed for MS17182 terminal lug.
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. The following list of hardware shall be packaged in a ventilated envelope made of material having a PH value of 6-8 and compatable with the weight of the contents. The packaged hardware shall be placed in the unit resolver package as follows:

Machine screw	(MS35275-213)	12 required (*) (**)
Lockwasher	(MS35338-135)	12 required (*) (**)
Drive washer	(MS17186-8)	1 required (*)
Nut	(MS17187-3)	1 required (*)

(*) For units with spline shaft and terminal board termination.
(**) For units with plain smooth shaft and terminal boatd termination.
For information only: Use mounting clamp MS17183 or equal.

7. Mark on housing is to match shaft mark within 10° when rotor is set at approx. resolver zero, and will bear no relation to the center lines shown on the drawing.
8. A voltage value of 1 to 130 V shall be entered in the 'Volts' designation space on the identification plate.
9. With the resolver energized under standard test conditions the resonant frequency will be recorded during qualification as that point where the output voltage has a maximum value and phase shift -90°. Low corner frequency (referenced as design information) is the point where output voltage is down 3db and phase shift 45°.
10. Pending issuance of a qualified products list (QPL) a sample of the resolvers to be furnished under this specification shall be subjected to the preproduction tests as specified in MIL-R-23417.

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TABLE I

RESOLVER, ELECTRICAL: TYPE 23R32W4a

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

	Requirements	Value	Unit	Tolerance
1.	Excitation winding	Stator		
2.	Rated frequency	400	Hz	±1%
3.	Input voltage		volts (rms)	±1%
	Maximum	130		
	Test	60		
	Minimum	1.0		
4.	Primary current	19	milliamps	nominal
5.	Primary power	0.078	watts	nominal
6.	D. C. resistance		ohms	±15%
	Rotor	172		
	Stator	97		
	Compensator	200		
7.	Impedance		ohms	R±15%+jx±10%
	Z_{RO}	296+j3170		
	Z_{SO}	216+j3108		
	Z_{CO}	322+j3013		
	Z_{RS}	244+j250		nominal
8.	Transformation ratio			
	Output/input	0.980		±1%
	Rotor/compensator	0.980		±0.1%
9.	T. R. variation (R/C)		%	
	Due to voltage	0.05		maximum
10.	Equality of transformation ratios		% spread	
	Output/input	0.1		maximum
	Rotor/compensator	0.1		maximum
11.	Phase shift		degrees	
	Output/input	1.9		±0.5
	Rotor/compensator	1.9		±0.5
12.	Phase shift variation (R/C)		degrees	
	Due to voltage	±0.2		
13.	Interaxis error	± 3	minutes	
14.	Functional error	±0.05	%	
15.	Null voltage		mV(rms) output	maximum
	Fundamental	20		
	Total	30		
16.	Shift of resolver zero		minutes	
	With voltage	±1.0		
	With frequency	±0.5		
17.	Harmonic distortion	0.1	%	maximum
18.	Temperature Rise	20	°C	maximum
19.	Friction torque	0.2	oz. - in.	maximum
	@ -55°C	0.4		maximum

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TABLE I (Continued)

Requirements	Value	Unit	Tolerance
20. Radial play	0.0010 (for reversal of 1/2 lb load)	inches	maximum
21. End play	0.0030 (for reversal of 1 lb load)	inches	±0.0020
22. Resonant frequency (see note 9)		kHz	minimum
23. Low corner frequency (see note 9)	12.6	Hz	maximum

Part number: The part number consists of the basic number of this specification sheet and a dash number taken from Table II.

Example:

M23417/17-01

Basic document number

Dash number taken from Table II

TABLE II

Dash No.	Shaft diameter	Shaft style	Termination	Shaft length	
				Free length - L (±.010)	Dim to stop on shaft - L' (±.010)
-01	.2405	Spline	Terminal block	.724	.676

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Custodians:
Army - MU
Navy - AS
Air Force - 11

Review activities:
Army - MU, EL
Navy - SH, AS, OS
Air Force - 11, 17, 85

User activities:
Army - MI, AV, ME, AT
Navy - CG, MC
Air Force - 14, 19

Preparing activity:
Army - MU

(Project 5990-0043)

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